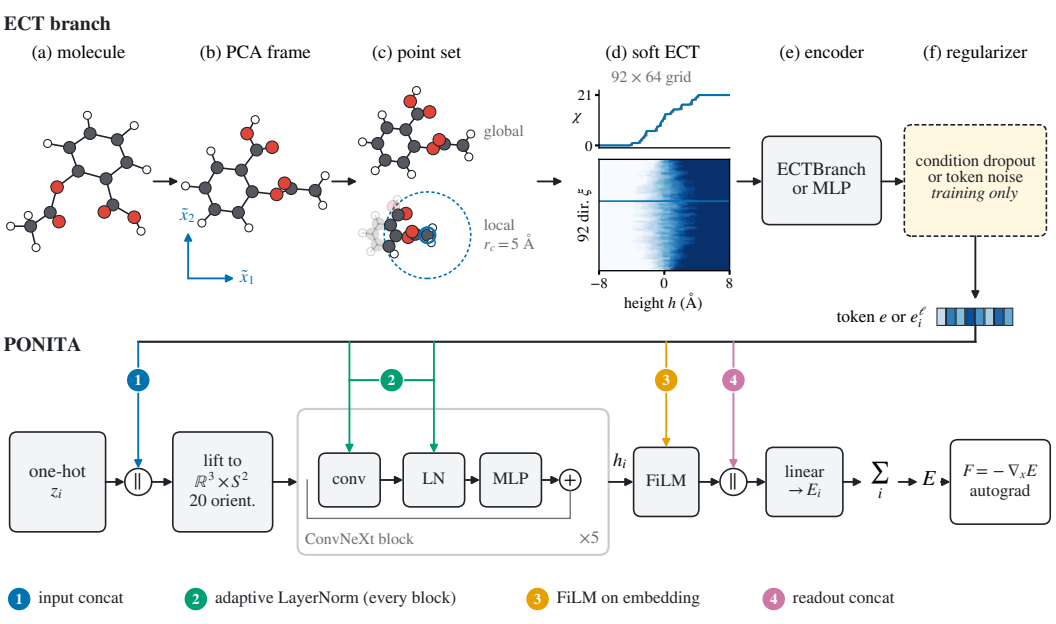
ECT branch
(a) molecule
(b) PCA frame
(c) point set
(d) soft ECT
(e) encoder
(f) regularizer
92 × 64 grid
21
χ
0
92 dir. ξ
−8
0
8
height h (Å)
global
local
r_c = 5 Å
x̃2
x̃1
ECTBranch or MLP
condition dropout or token noise
training only
token e or e_i^ℓ
PONITA
1
2
3
4
one-hot z_i
lift to ℝ³ × S² 20 orient.
conv
LN
MLP
ConvNeXt block
×5
h_i
FiLM
linear → E_i
Σ_i
E
F = −∇_x E autograd
1 input concat
2 adaptive LayerNorm (every block)
3 FiLM on embedding
4 readout concat